



SharePoint CAPSTONE PROJECT



Employee Directory



Capstone Project: Employee Directory Web Part

What is the Project

Build an Employee Directory Web Part using SPFx and SharePoint Lists

Project Overview

A company wants a simple way for employees to find colleagues without opening multiple systems.

Your task is to develop a custom **SharePoint Framework (SPFx) Web Part** that displays employee information from a SharePoint List.

This project is beginner-friendly and introduces students to:

-  SPFx Development
-  React Components
-  SharePoint Lists
-  REST API Calls
-  TypeScript
-  SharePoint Online

Project Objective

Create a custom SPFx web part called:

Employee Directory

The web part should display employees in card format and allow users to search for employees.

Business Requirements

Employees should be able to:

-  Search employees by name
-  View employee information

-  View department
 -  View email
 -  View phone number
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Project Architecture

SharePoint List



SPFx Web Part



React Component



Employee Cards

Step 1: Create SharePoint List

Create a list named:

Employee Directory

Columns

Column Name Type

Full Name	Single Line Text
Department	Choice
Email	Single Line Text
Phone	Single Line Text
Position	Single Line Text
Photo URL	Hyperlink

Sample Data

Full Name	Department	Position
John Smith	IT	Developer
Sarah Jones	HR	HR Manager
Michael Brown	Finance	Accountant
Emily Davis	Sales	Sales Executive

Step 2: Create SPFx Project

Create a new SPFx Web Part:

yo @microsoft/sharepoint

Project Settings

Project Name:
employee-directory

Framework:
React

Environment:
SharePoint Online

Component Type:
WebPart

Step 3: Design the UI

Each employee should appear as a card.

Example:

 John Smith

 Developer

 IT Department

 john@company.com

 0911000000

Step 4: Add Search Functionality

Create a search box.

Users should be able to:

- Search by name
- Search by department

Example:

Search: John

Results:

John Smith

Step 5: Read Data from SharePoint List

Use SharePoint REST API:

`/_api/web/lists/getbytitle('Employee Directory')/items`

Display

Show all employee records in cards.

Step 6: Add Department Filter

Create dropdown:

All Departments

IT

HR

Finance
Sales

Users should be able to filter employees.

Step 7: Responsive Design

The web part should work on:

 Desktop

 Mobile

 Tablet

Skills Learned

By completing this project, students will learn:

-  SPFx Fundamentals
-  React in SPFx
-  TypeScript
-  SharePoint Lists
-  SharePoint REST API
-  UI Development
-  SPFx Deployment

This is an ideal **first SPFx project** because it is small enough to finish in a few days while still teaching the core concepts every SharePoint developer needs.